

m -qh dgas: composition factors and shifted filtrations

Abstract

We define m -quasi-hereditary dg algebras by two requirements: the cohomology of each standard involves only simple modules with smaller or equal vertex labels in the chosen total order, with its own simple occurring exactly once in degree zero and never in negative degrees, and projective presentations have kernels filtered by allowed shifts of standards indexed by strictly larger vertices in that order. Exceptionality, fullness, and homological smoothness are consequences. We prove the equivalence with the exceptional-sequence formulation, show that the ordered projectives determine the standards and their left-dual sequence, and establish filtration criteria and the least possible value of m . Graded path algebras provide examples with arbitrary vertex orders and with non-trivial differentials. Counterexamples distinguish indispensable axioms, forbidden orders, insufficient values of m , and properties of classical filtrations that do not survive.

1 The two axioms

Fix an algebraically closed field $\mathbb{k}$ and integers $m, n \geq 1$. All dg modules are right modules; $\mathcal{D}(A)$ denotes their derived category. Complexes are cochain complexes: $d_X : X^q \rightarrow X^{q+1}$ and $H^q(X[s]) = H^{q+s}(X)$. Let A be a proper connective dg algebra which is m -truncated:

$$\dim_{\mathbb{k}} H^*(A) < \infty, \quad H^q(A) = 0 \quad (q \notin [1-m, 0]). \quad (1.1)$$

Assume that $B = H^0(A)$ is basic and that closed degree-zero orthogonal idempotents $e_1, \dots, e_n$ sum to 1, with classes forming a complete set of primitive idempotents of B . Write e_v also for its class in B , and let S_v be the simple top of $e_v B$. Set

$$\begin{aligned} P_v &= e_v A, & \mathcal{D}_{\text{fd}}(A) &= \{X \in \mathcal{D}(A) : \dim_{\mathbb{k}} H^*(X) < \infty\}, \\ \text{per } A &= \text{thick}_{\mathcal{D}(A)}(A), & \mathcal{H}_A^m &= \mathcal{D}_{\text{fd}}(A)^{[1-m, 0]}. \end{aligned} \quad (1.2)$$

Here thick denotes closure under shifts, cones, and direct summands; objects of $\text{per } A$ are called *perfect*. The interval in (1.2) specifies cohomological degrees. Write $\text{add } A$ for direct summands of finite direct sums of A . For a finite-dimensional B -module M , write $[M : S_w]$ for its composition multiplicity. Since B is split basic,

$$[M : S_w] = \dim_{\mathbb{k}} M e_w, \quad \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_w, X[t]) \cong H^t(X) e_w \quad (X \in \mathcal{D}_{\text{fd}}(A), t \in \mathbb{Z}). \quad (1.3)$$

Choose a permutation $\sigma \in S_n$ of the vertex labels. The vertex at position a is $\sigma(a)$; the position of vertex v is $\sigma^{-1}(v)$. Define

$$u \leq_{\sigma} v \iff \sigma^{-1}(u) \leq \sigma^{-1}(v), \quad u <_{\sigma} v \iff u \leq_{\sigma} v \text{ and } u \neq v. \quad (1.4)$$

The least and greatest vertices with respect to $\leq_{\sigma}$ are $\sigma(1)$ and $\sigma(n)$. Vertex labels and positions in a sequence are distinct.

The *m -extended module category* $\mathcal{H}_A^m$ is extriangulated: its conflations are the triangles $X \rightarrow Y \rightarrow Z \rightarrow X[1]$ with $X, Y, Z \in \mathcal{H}_A^m$. We write $X \rightarrow Y \rightarrow Z \dashrightarrow$ for such a conflation. The extension bifunctor and projective objects are

$$\mathbb{E}_{\mathcal{H}_A^m}(Z, X) = \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(Z, X[1]), \quad \mathcal{P}(\mathcal{H}_A^m) = \text{add } A; \quad (1.5)$$

see [MP25, §1 and Corollary 2.3]. For $\mathcal{S} \subseteq \mathcal{H}_A^m$, define $\text{Filt}_{\mathcal{H}_A^m}(\mathcal{S})$ using finite sequences of conflations

$$M_{a-1} \longrightarrow M_a \longrightarrow F_a \dashrightarrow, \quad M_0 = 0, \quad M_\ell = M, \quad F_a \in \mathcal{S} \quad (1 \leq a \leq \ell). \quad (1.6)$$

All intermediate objects M_a belong to $\mathcal{H}_A^m$. The empty filtration is permitted. Closure under direct summands is not part of the definition of $\text{Filt}_{\mathcal{H}_A^m}(\mathcal{S})$.

Definition 1.1 (m -qh dga). An algebra A satisfying the standing assumptions is an *m -qh dga with respect to σ* if there are objects $\Delta_v \in \mathcal{H}_A^m$, called *standard objects*, satisfying:

(dq1) For $1 \leq v, w \leq n$ and $q \in \mathbb{Z}$,

$$\begin{aligned} H^q(\Delta_v)e_w &= 0 \quad \text{if } v <_\sigma w, \\ H^q(\Delta_v)e_v &\cong \begin{cases} \mathbb{k}, & q = 0, \\ 0, & q \neq 0. \end{cases} \end{aligned} \quad (1.7)$$

(dq2) For every vertex v , there is a conflation

$$K_v \longrightarrow P_v \xrightarrow{p_v} \Delta_v \dashrightarrow \quad \text{in } \mathcal{H}_A^m, \quad (1.8)$$

with

$$\begin{aligned} K_v &\in \text{Filt}_{\mathcal{H}_A^m}(\mathcal{S}_{>v}^m), \\ \mathcal{S}_{>v}^m &= \{\Delta_w[s] : v <_\sigma w, 0 \leq s < m, \Delta_w[s] \in \mathcal{H}_A^m\}. \end{aligned} \quad (1.9)$$

We call the parameter m in Definition 1.1 the *width* of the structure. An integer $\ell \geq 1$ is an *admissible width* for (A, σ) if the standing assumptions and Definition 1.1 hold with $m = \ell$.

Condition dq1 says that $H^0(\Delta_v)$ has only composition factors S_w with $w \leq_\sigma v$, with S_v occurring once, whereas $H^q(\Delta_v)$ for $q < 0$ has only composition factors S_w with $w <_\sigma v$. Condition dq2 requires every shifted factor, as well as every intermediate object, to lie in $\mathcal{H}_A^m$. The object K_v is the first term of the conflation (1.8); no assertion that K_v is an ordinary module kernel is intended. For the greatest vertex $\sigma(n)$,

$$K_{\sigma(n)} = 0, \quad p_{\sigma(n)} : P_{\sigma(n)} \xrightarrow{\sim} \Delta_{\sigma(n)}. \quad (1.10)$$

Brickness, exceptionality, fullness, and smoothness are not additional requirements in Definition 1.1.

2 Perfection, fullness, and smoothness

Proposition 2.1 (The consequence of the filtration axiom alone). *For objects $\Delta_v \in \mathcal{H}_A^m$ satisfying condition dq2,*

$$\Delta_v \in \text{per } A \quad (1 \leq v \leq n), \quad \text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_1, \dots, \Delta_n) = \text{per } A. \quad (2.1)$$

For every vertex v one also has

$$\text{thick}_{\mathcal{D}_{\text{fd}}(A)}\{P_w : v <_\sigma w\} = \text{thick}_{\mathcal{D}_{\text{fd}}(A)}\{\Delta_w : v <_\sigma w\}. \quad (2.2)$$

Proof. Start with (1.10) and descend in $\leq_\sigma$. If Δ_w is perfect for all $w >_\sigma v$, the filtration (1.9) makes K_v perfect; the triangle (1.8) then makes Δ_v perfect. Conversely, (1.8) and (1.9) express every P_v in the thick subcategory generated by the standards. Restricting both arguments to vertices $w >_\sigma v$ proves (2.2). $\square$

Theorem 2.2 (Smoothness is a consequence). *Conditions [dq1](#) and [dq2](#) imply*

$$\begin{aligned} S_v &\in \text{per } A \quad (1 \leq v \leq n), \\ \mathcal{D}_{\text{fd}}(A) &= \text{per } A = \text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_1, \dots, \Delta_n), \\ A &\in \text{per}(A^{\text{op}} \otimes_{\mathbb{k}} A). \end{aligned} \quad (2.3)$$

The condition $A \in \text{per}(A^{\text{op}} \otimes_{\mathbb{k}} A)$ defines *homological smoothness*.

Proof. Step 1: perfect standards. Proposition [2.1](#) gives $\Delta_v \in \text{per } A$.

Step 2: perfect simples. Since $H^1(K_v) = 0$, the morphism $H^0(p_v) : e_v B \rightarrow H^0(\Delta_v)$ is surjective. Condition [dq1](#) gives $[H^0(\Delta_v) : S_v] = 1$. Thus $H^0(\Delta_v)$ has simple top S_v . Choose a top quotient $q_v : H^0(\Delta_v) \rightarrow S_v$, and let $a_v : \Delta_v \rightarrow S_v$ be the composite of $\Delta_v \rightarrow H^0(\Delta_v)$ with q_v . Take the triangle

$$T_v \longrightarrow \Delta_v \xrightarrow{a_v} S_v \longrightarrow T_v[1]. \quad (2.4)$$

The cohomology sequence of [\(2.4\)](#) gives

$$H^q(T_v) \cong \begin{cases} H^q(\Delta_v), & q < 0, \\ \text{Ker } q_v, & q = 0, \\ 0, & q > 0. \end{cases} \quad (2.5)$$

Every composition factor of every $H^q(T_v)$ is S_w with $w <_{\sigma} v$, by condition [dq1](#). Finite truncation triangles and composition series give

$$T_v \in \text{thick}_{\mathcal{D}_{\text{fd}}(A)}\{S_w : w <_{\sigma} v\}. \quad (2.6)$$

For the least vertex $\sigma(1)$, $T_{\sigma(1)} = 0$ and $S_{\sigma(1)} \cong \Delta_{\sigma(1)}$ is perfect. Ascending induction using [\(2.4\)](#) makes every S_v perfect.

Step 3: fullness and smoothness. Every object of $\mathcal{D}_{\text{fd}}(A)$ is obtained from the S_v by finite truncation triangles and composition series. Consequently $\mathcal{D}_{\text{fd}}(A) \subseteq \text{per } A$. Properness gives the reverse inclusion. Finally, $B/\text{rad } B \cong \mathbb{k}^n$ is separable over $\mathbb{k}$ and perfect over A . The criterion [[RS22](#), Theorem 5.8] proves $A \in \text{per}(A^{\text{op}} \otimes_{\mathbb{k}} A)$. $\square$

3 Exceptionality and finite Hom tests

An object $E \in \mathcal{D}_{\text{fd}}(A)$ is *exceptional* if

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(E, E[t]) \cong \begin{cases} \mathbb{k}, & t = 0, \\ 0, & t \neq 0 \end{cases} \quad (t \in \mathbb{Z}). \quad (3.1)$$

An ordered tuple $(E_1, \dots, E_n)$ is an *exceptional sequence* if every E_a is exceptional and

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(E_a, E_b[t]) = 0 \quad (1 \leq b < a \leq n, t \in \mathbb{Z}). \quad (3.2)$$

The sequence is *full* if its terms thickly generate $\mathcal{D}_{\text{fd}}(A)$. We use the ordering

$$\Delta^{\sigma} = (\Delta_{\sigma(1)}, \dots, \Delta_{\sigma(n)}). \quad (3.3)$$

Thus standards indexed by larger vertices in $\leq_{\sigma}$ occur later in [\(3.3\)](#), and backward orthogonality means vanishing from Δ_w to $\Delta_v[t]$ for $w >_{\sigma} v$.

Lemma 3.1 (Orthogonality from composition factors). *Assume condition [dq2](#) and*

$$H^t(\Delta_v)e_w = 0 \quad (v <_{\sigma} w, t \in \mathbb{Z}). \quad (3.4)$$

Then

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_w, \Delta_v[t]) = 0 \quad (v <_{\sigma} w, t \in \mathbb{Z}), \quad (3.5)$$

$$p_v^* : \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_v, \Delta_v[t]) \xrightarrow{\sim} \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_v, \Delta_v[t]) \cong H^t(\Delta_v)e_v \quad (1 \leq v \leq n, t \in \mathbb{Z}). \quad (3.6)$$

Proof. Fix v and consider the vertices $w >_\sigma v$ in decreasing order with respect to $\leq_\sigma$. Equation (1.3) and (3.4) give $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_w, \Delta_v[t]) = 0$ for every t . Once every standard Δ_u with $u >_\sigma w$ is orthogonal to Δ_v in every degree, the filtration of K_w in (1.9) gives the same vanishing for K_w . Apply $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(-, \Delta_v[t])$ to the presentation (1.8) for w to obtain (3.5). The filtration of K_v now implies $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(K_v, \Delta_v[t]) = 0$ for every t . The presentation for v gives (3.6). $\square$

Theorem 3.2 (Equivalent formulations under the filtration axiom). *For objects $\Delta_v \in \mathcal{H}_A^m$ satisfying condition dq2, the following are equivalent:*

- (i) condition dq1;
- (ii) Δ^σ is an exceptional sequence;
- (iii) Δ^σ is full exceptional in $\mathcal{D}_{\text{fd}}(A) = \text{per } A$.

In particular, $\text{End}_{\mathcal{H}_A^m}(\Delta_v) \cong \mathbb{k}$ is a consequence of conditions dq1 and dq2.

Proof. We prove (i) $\Rightarrow$ (ii) $\Rightarrow$ (i), then (i) $\Rightarrow$ (iii); (iii) $\Rightarrow$ (ii) follows from the definition.

(i) $\Rightarrow$ (ii). Lemma 3.1 gives (3.5). Equations (3.6) and (1.7) give (3.1) for every Δ_v .

(ii) $\Rightarrow$ (i). For $w >_\sigma v$, equation (2.2) places P_w in the thick subcategory generated by Δ_u with $u >_\sigma v$. Equation (3.2) gives $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_w, \Delta_v[t]) = 0$ for all t . Thus (3.4) holds. The isomorphism (3.6) and exceptionality of Δ_v give $H^t(\Delta_v)e_v \cong \mathbb{k}$ for $t = 0$ and zero for $t \neq 0$, completing (1.7).

(i) $\Rightarrow$ (iii). Theorem 2.2 gives fullness and $\mathcal{D}_{\text{fd}}(A) = \text{per } A$; the implication (i) $\Rightarrow$ (ii) gives exceptionality. $\square$

Proposition 3.3 (Tests inside the extended module category). *For every vertex w and $0 \leq s < m$, define*

$$U_{w,s} = \tau^{\geq 1-m}(P_w[s]) \in \mathcal{H}_A^m. \quad (3.7)$$

There are natural isomorphisms

$$\text{Hom}_{\mathcal{H}_A^m}(U_{w,s}, \Delta_v) \cong \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_w, \Delta_v[-s]) \cong H^{-s}(\Delta_v)e_w. \quad (3.8)$$

Consequently condition dq1 is equivalent to

$$\begin{aligned} \text{Hom}_{\mathcal{H}_A^m}(U_{w,s}, \Delta_v) &= 0 \quad (v <_\sigma w, 0 \leq s < m), \\ \text{Hom}_{\mathcal{H}_A^m}(U_{v,s}, \Delta_v) &= 0 \quad (1 \leq s < m), \\ \dim_{\mathbb{k}} \text{Hom}_{\mathcal{H}_A^m}(P_v, \Delta_v) &= 1 \quad (1 \leq v \leq n). \end{aligned} \quad (3.9)$$

Under condition dq2 and the first line of (3.9), the last line can equivalently be replaced by the brick condition $\text{End}_{\mathcal{H}_A^m}(\Delta_v) \cong \mathbb{k}$ for every v .

Proof. Put $L_{w,s} = \tau^{\leq -m}(P_w[s])$. In the triangle

$$L_{w,s} \longrightarrow P_w[s] \longrightarrow U_{w,s} \longrightarrow L_{w,s}[1],$$

both $L_{w,s}$ and $L_{w,s}[1]$ lie in $\mathcal{D}_{\text{fd}}(A)^{\leq -m}$, whereas Δ_v lies in $\mathcal{D}_{\text{fd}}(A)^{\geq 1-m}$. Applying $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(-, \Delta_v)$ proves (3.8). Since all other cohomological degrees of Δ_v vanish, (3.9) is equivalent to (1.7). Under condition dq2, (3.6) identifies the last vector space in (3.9) with $\text{End}_{\mathcal{H}_A^m}(\Delta_v)$. $\square$

Corollary 3.4 (Classical specialisation). *For $m = 1$, Definition 1.1 is equivalent to classical quasi-heredity of $B = H^0(A)$ with respect to $\leq_\sigma$.*

Proof. For $m = 1$, the dga A is quasi-isomorphic to B , and $\mathcal{H}_A^1 \simeq \text{mod } B$. We prove both implications.

Conditions **dq1** and **dq2** $\Rightarrow$ classical quasi-heredity. Only the shift $s = 0$ is allowed in (1.9), and (1.8) becomes $0 \rightarrow K_v \rightarrow e_v B \rightarrow \Delta_v \rightarrow 0$. Condition **dq1** gives composition factors S_w with $w \leq_\sigma v$, with S_v occurring once. These are the classical projective presentation axioms [BB26, §2.1].

Classical quasi-heredity $\Rightarrow$ conditions **dq1** and **dq2**. The classical standard modules have the required composition factors, and the projective kernel K_v is filtered by Δ_w with $w >_\sigma v$. Thus (1.7) and (1.9) hold with $s = 0$. $\square$

4 Canonical standards and costandards

Assume conditions **dq1** and **dq2** throughout Sections 4–6.

Proposition 4.1 (Standards determined by the ordered projectives). *For a vertex v , set*

$$\mathcal{Q}_v = \text{thick}_{\mathcal{D}_{\text{fd}}(A)}\{P_w : v <_\sigma w\}, \quad e_{>v}^\sigma = \sum_{v <_\sigma w} e_w. \quad (4.1)$$

Define $\mathcal{Q}_v^\perp$ to consist of the objects $X \in \mathcal{D}_{\text{fd}}(A)$ satisfying $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(Y, X[t]) = 0$ for every $Y \in \mathcal{Q}_v$ and $t \in \mathbb{Z}$. Then

$$\begin{aligned} K_v &\in \mathcal{Q}_v, & \Delta_v &\in \mathcal{Q}_v^\perp, \\ p_v^* : \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_v, X[t]) &\xrightarrow{\sim} \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_v, X[t]) \quad (X \in \mathcal{Q}_v^\perp, t \in \mathbb{Z}). \end{aligned} \quad (4.2)$$

The pair (Δ_v, p_v) is determined by (A, σ) up to a unique isomorphism compatible with p_v , and K_v is determined up to isomorphism. Moreover,

$$H^0(\Delta_v) \cong e_v B / e_v B e_{>v}^\sigma B, \quad B = H^0(A). \quad (4.3)$$

The morphism $H^0(p_v) : e_v B \rightarrow H^0(\Delta_v)$ is a projective cover with simple top S_v .

Proof. Proposition 2.1 identifies $\mathcal{Q}_v$ with $\text{thick}_{\mathcal{D}_{\text{fd}}(A)}\{\Delta_w : v <_\sigma w\}$. The filtration of K_v therefore gives $K_v \in \mathcal{Q}_v$. Condition **dq1** gives $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_w, \Delta_v[t]) = H^t(\Delta_v)e_w = 0$ for $w >_\sigma v$, so $\Delta_v \in \mathcal{Q}_v^\perp$. Applying $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(-, X[t])$ to the presentation triangle (1.8) proves (4.2). The universal property (4.2) proves uniqueness of (Δ_v, p_v) ; cones then give uniqueness of K_v up to isomorphism.

Since $H^1(K_v) = 0$, the morphism $H^0(p_v)$ is surjective. Put $D_v^0 = e_v B / e_v B e_{>v}^\sigma B$. The support condition in **dq1** makes $H^0(p_v)$ factor through a surjection $f_v : D_v^0 \rightarrow H^0(\Delta_v)$. The module D_v^0 belongs to $\mathcal{Q}_v^\perp$, so its quotient morphism $P_v \rightarrow D_v^0$ factors uniquely through p_v . Taking degree-zero cohomology gives $g_v : H^0(\Delta_v) \rightarrow D_v^0$. Precomposition with the surjections from $e_v B$ shows $g_v f_v = 1_{D_v^0}$ and $f_v g_v = 1_{H^0(\Delta_v)}$. This proves (4.3). Condition **dq1** gives $H^0(\Delta_v)e_v \cong \mathbb{k}$, so $H^0(\Delta_v)$ is a non-zero quotient of the indecomposable projective $e_v B$. The top of $H^0(\Delta_v)$ is S_v , and $H^0(p_v) : e_v B \rightarrow H^0(\Delta_v)$ is a projective cover. $\square$

Definition 4.2 (Costandard sequence). The *left-dual sequence* of Δ^σ is the exceptional sequence $(\nabla_{\sigma(n)}, \dots, \nabla_{\sigma(1)})$ satisfying

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_u, \nabla_v[t]) \cong \begin{cases} \mathbb{k}, & u = v \text{ and } t = 0, \\ 0, & \text{otherwise} \end{cases} \quad (1 \leq u, v \leq n, t \in \mathbb{Z}). \quad (4.4)$$

The objects ∇_v are the *costandards*. The ordering and pairing in Definition 4.2 specify the convention for left-dual sequences; compare [BB26, §2.3].

Proposition 4.3 (Existence, support, and amplitude of costandards). *The left-dual sequence in Definition 4.2 exists and is unique up to isomorphism of its terms. It is full in $\mathcal{D}_{\text{fd}}(A) = \text{per } A$, and*

$$\nabla_v \in \mathcal{H}_A^m[1-m] = \mathcal{D}_{\text{fd}}(A)^{[0, m-1]}. \quad (4.5)$$

The vertex components of each ∇_v satisfy

$$\begin{aligned} H^t(\nabla_v)e_w &= 0 \quad (v <_\sigma w, t \in \mathbb{Z}), \\ H^t(\nabla_v)e_v &\cong \begin{cases} \mathbb{k}, & t = 0, \\ 0, & t \neq 0. \end{cases} \end{aligned} \quad (4.6)$$

Proof. Step 1: construction and uniqueness. For $v = \sigma(a)$, start with $Y_a = \Delta_v$. For $j = a-1, a-2, \dots, 1$, form evaluation triangles

$$\begin{aligned} V_j &= \bigoplus_{t \in \mathbb{Z}} \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_{\sigma(j)}, Y_{j+1}[t])[-t], \\ V_j \otimes_{\mathbb{k}} \Delta_{\sigma(j)} &\longrightarrow Y_{j+1} \longrightarrow Y_j \longrightarrow (V_j \otimes_{\mathbb{k}} \Delta_{\sigma(j)})[1]. \end{aligned} \quad (4.7)$$

The complexes V_j have zero differential and finite total dimension: all standards are perfect, and every Y_{j+1} has finite-dimensional total cohomology. Put $\nabla_v = Y_1$; for $a = 1$, put $\nabla_v = \Delta_v$. Exceptionality annihilates morphisms from $\Delta_{\sigma(j)}$ at step j , and backward orthogonality preserves the vanishings already obtained. Thus ∇_v lies in $\text{thick}_{\mathcal{D}_{\text{fd}}(A)}\{\Delta_u : u \leq_\sigma v\}$ and is right orthogonal to every Δ_u with $u <_\sigma v$. The same triangles give (4.4). Combining the mutation triangles gives a triangle $M_v \rightarrow \Delta_v \rightarrow \nabla_v \rightarrow M_v[1]$, with $M_v \in \text{thick}_{\mathcal{D}_{\text{fd}}(A)}\{\Delta_u : u <_\sigma v\}$. Consequently

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\nabla_v, \nabla_v[t]) \cong \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_v, \nabla_v[t]),$$

so ∇_v is exceptional. For $u <_\sigma v$, the object ∇_u lies in the subcategory generated by Δ_z with $z <_\sigma v$, giving $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\nabla_u, \nabla_v[t]) = 0$. Induction in a using the triangles for M_v shows that the costandards generate every standard, hence gives fullness of the reversed sequence.

For uniqueness, an object $N_v \in \mathcal{D}_{\text{fd}}(A)$ satisfying (4.4) belongs to $\text{thick}_{\mathcal{D}_{\text{fd}}(A)}\{\Delta_u : u \leq_\sigma v\}$: evaluation against Δ^σ has no factors belonging to $\text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_w)$ with $w >_\sigma v$. A generator of $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_v, N_v) \cong \mathbb{k}$ gives a triangle $M_v \rightarrow \Delta_v \rightarrow N_v \rightarrow M_v[1]$, where $M_v \in \text{thick}_{\mathcal{D}_{\text{fd}}(A)}\{\Delta_u : u <_\sigma v\}$. Since N_v is right orthogonal to that subcategory, its localisation property identifies N_v with the object constructed in (4.7).

Step 2: the cohomological window. For any allowed kernel factor, (4.4) gives

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_w[s], \nabla_u[t]) \cong \begin{cases} \mathbb{k}, & w = u \text{ and } t = s, \\ 0, & \text{otherwise.} \end{cases} \quad (4.8)$$

Since $0 \leq s < m$, the filtration of K_v implies $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(K_v, \nabla_u[t]) = 0$ for $t \notin [0, m-1]$. Apply $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(-, \nabla_u[t])$ to (1.8); equation (4.4) then gives $H^t(\nabla_v)e_v = 0$ for every vertex v and $t \notin [0, m-1]$. This proves (4.5).

Step 3: vertex support. The factors in a standard filtration of P_w , supplied by condition dq2, are shifts of Δ_z with $z \geq_\sigma w$. If $w >_\sigma v$, all their morphisms to shifts of ∇_v vanish by (4.8). If $w = v$, only the quotient Δ_v contributes, in degree zero. Since $H^t(\nabla_v)e_w = \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_w, \nabla_v[t])$, the calculations for $w >_\sigma v$ and $w = v$ prove (4.6). $\square$

5 Filtrations and the least possible width

For $\ell \geq 1$, define

$$\begin{aligned} I_w^\ell &= \{s \in \mathbb{Z} : 0 \leq s < \ell, \Delta_w[s] \in \mathcal{H}_A^\ell\}, \\ \mathcal{S}_{>v}^\ell &= \{\Delta_w[s] : v <_\sigma w, s \in I_w^\ell\}. \end{aligned} \quad (5.1)$$

Here $\mathcal{H}_A^\ell = \mathcal{D}_{\text{fd}}(A)^{[1-\ell, 0]}$, whether or not A has cohomology concentrated in that interval.

Proposition 5.1 (Recognition of shifted standard filtrations). *For $X \in \mathcal{Q}_v$, the following conditions are equivalent:*

- (1) $X \in \text{Filt}_{\mathcal{H}_A^\ell}(\mathcal{S}_{>v}^\ell)$;
- (2) $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(X, \nabla_w[t]) = 0$ for every $w >_\sigma v$ and $t \notin I_w^\ell$.

If condition (2) holds, put $a = \sigma^{-1}(v)$. The object X admits a filtration with successive factors $G_{\sigma(n)}(X), \dots, G_{\sigma(a+1)}(X)$, where

$$G_w(X) \cong \bigoplus_{s \in I_w^\ell} \Delta_w[s]^{\oplus \dim_{\mathbb{k}} \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(X, \nabla_w[s])}. \quad (5.2)$$

In particular, $\text{Filt}_{\mathcal{H}_A^\ell}(\mathcal{S}_{>v}^\ell)$ is closed under direct summands.

Proof. (1) $\Rightarrow$ (2). Every allowed factor satisfies the required vanishings by (4.8); the long exact sequences of the filtration preserve the vanishings in condition (2).

(2) $\Rightarrow$ (1). Put $a = \sigma^{-1}(v)$. The full exceptional sequence in $\mathcal{Q}_v$ is $(\Delta_{\sigma(a+1)}, \dots, \Delta_{\sigma(n)})$. Successive evaluation against $\Delta_{\sigma(n)}, \dots, \Delta_{\sigma(a+1)}$ gives a tower with one factor $G_w(X) \in \text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_w)$ for each $w >_\sigma v$. Equivalently, the octahedral axiom gives a finite sequence of triangles starting at zero and ending at X , with successive factors $G_{\sigma(n)}(X), \dots, G_{\sigma(a+1)}(X)$. Exceptionality expresses $G_w(X)$ as a finite direct sum of shifts of Δ_w . For a fixed w , applying $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(-, \nabla_w[t])$ to the tower annihilates every factor $G_u(X)$ with $u \neq w$. Consequently the multiplicity of $\Delta_w[t]$ in $G_w(X)$ is $\dim_{\mathbb{k}} \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(X, \nabla_w[t])$. Condition (2) gives (5.2). Every factor in (5.2) belongs to $\mathcal{H}_A^\ell$; extension closure puts every intermediate object in $\mathcal{H}_A^\ell$. Refining the finite direct sums gives condition (1). When $v = \sigma(n)$, the category $\mathcal{Q}_v$ is zero, and the empty filtration proves the assertion.

Both $\mathcal{Q}_v$ and the vanishings in condition (2) are closed under direct summands, proving closure of $\text{Filt}_{\mathcal{H}_A^\ell}(\mathcal{S}_{>v}^\ell)$ under direct summands. $\square$

For vertices v, w and $s \in \mathbb{Z}$, set

$$b_{vw,s} = \dim_{\mathbb{k}} \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_v, \nabla_w[s]) = \dim_{\mathbb{k}} H^s(\nabla_w)e_v. \quad (5.3)$$

For $v <_\sigma w$, applying $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(-, \nabla_w[s])$ to (1.8) gives

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(K_v, \nabla_w[s]) \cong \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_v, \nabla_w[s]). \quad (5.4)$$

Thus the ordered filtration of K_v in Proposition 5.1 contains exactly $b_{vw,s}$ copies of $\Delta_w[s]$ in the factor $G_w(K_v)$.

Example 5.2 (Unsigned filtration multiplicities are not invariant). Suppose $w >_\sigma v$ and both $\Delta_w[s]$ and $\Delta_w[s+1]$ belong to $\mathcal{S}_{>v}^m$. The triangles

$$0 \longrightarrow \Delta_w[s] \xrightarrow{1} \Delta_w[s] \longrightarrow 0, \quad \Delta_w[s] \longrightarrow 0 \longrightarrow \Delta_w[s+1] \xrightarrow{1} \Delta_w[s+1] \quad (5.5)$$

define a two-factor filtration of the zero object, with factors $\Delta_w[s]$ and $\Delta_w[s+1]$. For a concrete instance, take $A = \mathbb{k} \times \mathbb{k}$, $m = 2$, $1 <_\sigma 2$, $\Delta_i = P_i = S_i$, $w = 2$, and $s = 0$. Therefore arbitrary filtrations need not have invariant lengths or invariant numbers of shifted factors, even for a semisimple algebra. The multiplicities in the ordered factors (5.2) are nevertheless invariant.

Corollary 5.3 (Least width). Suppose (A, σ) admits conditions dq1 and dq2 for at least one width. For its canonical standards and costandards, put

$$\begin{aligned} \alpha_A &= -\min\{q : H^q(A) \neq 0\}, \\ \alpha_w &= -\min\{q : H^q(\Delta_w) \neq 0\}, \quad \beta_w = \max\{q : H^q(\nabla_w) \neq 0\}. \end{aligned} \quad (5.6)$$

These integers are non-negative. The least admissible width is

$$m_{\min}(A, \sigma) = 1 + \max \left\{ \alpha_A, \max_{1 \leq w \leq n} (\alpha_w + \beta_w) \right\}. \quad (5.7)$$

Every width $m \geq m_{\min}(A, \sigma)$ is admissible.

Proof. Propositions 4.1 and 4.3 make the objects in (5.6) independent of the chosen width. $H^0(A)$ is non-zero. Condition dq1 and (4.6) give $H^0(\Delta_w)e_w \cong H^0(\nabla_w)e_w \cong \mathbb{k}$ for every vertex w . Hence all the integers in (5.6) are non-negative. For $s \geq 0$, the cohomological shift convention gives

$$\Delta_w[s] \in \mathcal{H}_A^\ell \iff 0 \leq s \leq \ell - 1 - \alpha_w. \quad (5.8)$$

If $\beta_w > 0$, equation (4.6) gives a vertex $v <_\sigma w$ with $b_{vw, \beta_w} > 0$. Necessity in Proposition 5.1, applied to K_v , then forces $\alpha_w + \beta_w \leq \ell - 1$ for every admissible width ℓ . If $\beta_w = 0$, the same inequality follows from $\Delta_w \in \mathcal{H}_A^\ell$. Also $\alpha_A \leq \ell - 1$ is necessary.

Conversely, assume $\alpha_A \leq \ell - 1$ and $\alpha_w + \beta_w \leq \ell - 1$ for every w . The objects P_v and Δ_v lie in $\mathcal{H}_A^\ell$. Every shift occurring in the ordered factors $G_w(K_v)$ satisfies $s \leq \beta_w$; equation (5.8) therefore puts each factor in $\mathcal{H}_A^\ell$. Proposition 5.1 gives the required kernel filtration, including $K_v \in \mathcal{H}_A^\ell$. The presentation triangle is a conflation in $\mathcal{H}_A^\ell$, and condition dq1 is unchanged. This proves (5.7) and the assertion about larger widths. $\square$

Corollary 5.4 (Standards and costandards concentrated in degree zero). *The following are equivalent:*

- (i) $\Delta_v, \nabla_v \in \text{mod } B$ for every vertex v ;
- (ii) $m_{\min}(A, \sigma) = 1$.

In either case, A is quasi-isomorphic to the ordinary quasi-hereditary algebra B with the given order.

Proof. (i) $\Rightarrow$ (ii). Degree-zero concentration of every ∇_w gives $b_{vw, s} = 0$ for $s \neq 0$ in (5.3). Proposition 5.1 therefore gives unshifted standard filtrations of the kernels K_v . Since every Δ_w is a module, every K_v and then every P_v belongs to $\text{mod } B$. Thus $A \simeq B$ and all three integers in (5.6) are zero. Equation (5.7) gives $m_{\min}(A, \sigma) = 1$.

(ii) $\Rightarrow$ (i). Apply Definition 1.1 and Proposition 4.3 at width 1. Corollary 3.4 gives classical quasi-heredity. $\square$

6 Grothendieck groups, injectives, and opposite algebras

The *Grothendieck group* $K_0(\mathcal{D}_{\text{fd}}(A))$ is generated by symbols $[X]$, with relations $[Y] = [X] + [Z]$ for triangles $X \rightarrow Y \rightarrow Z \rightarrow X[1]$. Since $\mathcal{D}_{\text{fd}}(A) = \text{per } A$ and A is proper, the *Euler pairing*

$$\chi_A(X, Y) = \sum_{t \in \mathbb{Z}} (-1)^t \dim_{\mathbb{k}} \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(X, Y[t]) \quad (6.1)$$

is a finite sum and induces a bilinear form on $K_0(\mathcal{D}_{\text{fd}}(A))$.

Proposition 6.1 (Bases, multiplicities, and a determinant). *The classes of the simples, standards, costandards, and projectives are four $\mathbb{Z}$ -bases of $K_0(\mathcal{D}_{\text{fd}}(A)) \cong \mathbb{Z}^n$. The standard and costandard bases satisfy*

$$\chi_A(\Delta_v, \nabla_w) = \delta_{vw}. \quad (6.2)$$

In the projective expansion in the standard basis,

$$[P_v] = [\Delta_v] + \sum_{w >_\sigma v} \left(\sum_{s=0}^{m-1} (-1)^s b_{vw, s} \right) [\Delta_w]. \quad (6.3)$$

If $c_{vw,s}$ counts factors $\Delta_w[s]$ in any filtration of K_v , then

$$\sum_{s=0}^{m-1} (-1)^s c_{vw,s} = \sum_{s=0}^{m-1} (-1)^s b_{vw,s} \quad (v <_\sigma w). \quad (6.4)$$

Finally, define the *Euler Cartan matrix* by

$$C_{vu} = \sum_{q \in \mathbb{Z}} (-1)^q [H^q(P_v) : S_u] = \chi_A(P_u, P_v). \quad (6.5)$$

Then $\det C = 1$.

Proof. The bounded standard t -structure identifies $K_0(\mathcal{D}_{\text{fd}}(A))$ with $K_0(\text{mod } B)$, with simple basis $[S_v]$. Condition **dq1** gives $[\Delta_v] = [S_v] + \sum_{u <_\sigma v} d_{vu} [S_u]$ for integers d_{vu} , so the standard classes are a basis. Equation (4.6) proves the analogous assertion for costandard classes. Equation (4.4) gives (6.2). The ordered filtration in Proposition 5.1 gives (6.3); its transition matrix is upper unitriangular when rows and columns are listed in σ -order. Thus the projective classes form a basis as well. An arbitrary kernel filtration and the relation $[\Delta_w[s]] = (-1)^s [\Delta_w]$ prove (6.4) by comparison in the standard basis. The transition matrix from standards to simples is lower unitriangular. The product of the projective-to-standard and standard-to-simple transition matrices is C , proving $\det C = 1$. $\square$

The integers in (6.3) and the entries of (6.5) can be negative. Proposition 6.1 concerns the Euler Cartan matrix of A ; it does not assert that the ordinary Cartan matrix of $H^0(A)$ has determinant 1.

For a complex V of $\mathbb{k}$ -vector spaces, write $\mathbb{D}V = \text{Hom}_{\mathbb{k}}^\bullet(V, \mathbb{k})$ for its complex dual. Vector-space duality exchanges right A -modules and right A^{op} -modules and satisfies $H^q(\mathbb{D}X) \cong \mathbb{D}H^{-q}(X)$.

Proposition 6.2 (Injective presentations and opposite algebras). *Put $\widehat{\nabla}_v = \nabla_v[m-1]$ and $I_v = \mathbb{D}(Ae_v)[m-1]$. There are conflations*

$$\begin{aligned} \widehat{\nabla}_v &\longrightarrow I_v \longrightarrow L_v \dashrightarrow \quad \text{in } \mathcal{H}_A^m, \\ L_v &\in \text{Filt}_{\mathcal{H}_A^m} \{ \widehat{\nabla}_w[-s] : v <_\sigma w, \ 0 \leq s < m, \ \widehat{\nabla}_w[-s] \in \mathcal{H}_A^m \}. \end{aligned} \quad (6.6)$$

The objects I_v are the indecomposable injectives of $\mathcal{H}_A^m$. The dg algebra A^{op} satisfies conditions **dq1** and **dq2** for the same m and order, with standards $\mathbb{D}\nabla_v$ and costandards $\mathbb{D}\Delta_v$. In particular,

$$m_{\min}(A^{\text{op}}, \sigma) = m_{\min}(A, \sigma). \quad (6.7)$$

Proof. The shifted vector-space duality $X \mapsto \mathbb{D}X[m-1]$ is a contravariant equivalence $\mathcal{H}_A^m \rightarrow \mathcal{H}_{A^{\text{op}}}^m$, taking projectives to injectives. The projectives of $\mathcal{H}_{A^{\text{op}}}^m$ are the summands of A as a left A -module, so the corresponding right-module injectives are precisely the objects I_v .

Decompose I_v using the full exceptional sequence $(\nabla_{\sigma(n)}, \dots, \nabla_{\sigma(1)})$. The same evaluation-tower argument as in Proposition 5.1, now using $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_w, -)$, gives successive factors in $\text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\nabla_{\sigma(1)}), \dots, \text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\nabla_{\sigma(n)})$. The identity

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_w, I_v[t]) \cong \mathbb{D}(H^{1-m-t}(\Delta_w)e_v) \quad (6.8)$$

identifies the factor in $\text{thick}_{\mathcal{D}_{\text{fd}}(A)}(\nabla_w)$ as

$$\bigoplus_{s \geq 0} \nabla_w[m-1-s]^{\oplus \dim_{\mathbb{k}} H^{-s}(\Delta_w)e_v}. \quad (6.9)$$

For $w <_\sigma v$, condition **dq1** makes the factor (6.9) zero; for $w = v$, the factor is exactly $\widehat{\nabla}_v$. Every occurring s satisfies $s \leq \alpha_w$. Corollary 5.3 gives $\beta_w + s \leq \beta_w + \alpha_w \leq m-1$, so all factors (6.9) belong to $\mathcal{H}_A^m$. The tower with factors (6.9) therefore gives (6.6).

Apply $\mathbb{D}(-)[m-1]$ to the conflation in (6.6). The resulting conflation in $\mathcal{H}_{A^{\text{op}}}^m$ is

$$\mathbb{D}L_v[m-1] \longrightarrow Ae_v \longrightarrow \mathbb{D}\nabla_v \dashrightarrow, \quad (6.10)$$

and its kernel factors are $(\mathbb{D}\nabla_w)[s]$ with $w >_\sigma v$, $0 \leq s < m$, and $(\mathbb{D}\nabla_w)[s] \in \mathcal{H}_{A^{\text{op}}}^m$. The support formula (4.6), after taking vector-space duals, verifies condition dq1 for $\mathbb{D}\nabla_v$. Thus (6.10) proves condition dq2 for A^{op} . Finally,

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A^{\text{op}})}(\mathbb{D}\nabla_u, \mathbb{D}\Delta_v[t]) \cong \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_v, \nabla_u[t]) \quad (6.11)$$

identifies $\mathbb{D}\Delta_v$ as the costandards for A^{op} . Applying the result to both A and A^{op} proves (6.7). $\square$

7 Graded path dg algebras and explicit structures

Let Q be a finite acyclic quiver on $\{1, \dots, n\}$, with all arrow degrees at most zero. We concatenate paths from left to right: $e_u a = a = a e_v$ for an arrow $a : u \rightarrow v$. Let $A = (\mathbb{k}Q, d_A)$, where d_A is a graded derivation of degree 1, $d_A^2 = 0$, and $d_A(e_v) = 0$ for every vertex v . For a path p , write $|p|$ for its degree, with $|e_v| = 0$, and set

$$M = 1 - \min\{|p| : p \text{ is a path in } Q\}. \quad (7.1)$$

The integer M measures the degrees in the underlying graded algebra. The least integer m satisfying $H^q(A) = 0$ for $q \notin [1-m, 0]$ can be smaller than M when $d_A \neq 0$.

Theorem 7.1 (Every total order at the path-degree bound). *For every $\sigma \in S_n$, put*

$$e_{>v}^\sigma = \sum_{v <_\sigma w} e_w, \quad C_v = A/Ae_{>v}^\sigma A, \quad \Delta_v = e_v C_v, \quad K_v = e_v A e_{>v}^\sigma A. \quad (7.2)$$

The objects in (7.2), with their quotient maps $P_v \twoheadrightarrow \Delta_v$, satisfy conditions dq1 and dq2 for every $m \geq M$.

Proof. Step 1: condition dq1. The module Δ_v has a basis consisting of paths starting at v whose vertices u all satisfy $u \leq_\sigma v$. Consequently $\Delta_v e_w = 0$ for $v <_\sigma w$. Acyclicity gives $\Delta_v e_v = \mathbb{k}e_v$ in degree zero. These equalities of complexes imply condition dq1.

Step 2: the kernel filtration. Fix $v = \sigma(i)$. If $i = n$, then $K_v = 0$ and its filtration is empty. For $i < n$, define dg ideals and dg submodules by

$$\begin{aligned} J_j &= A \left(\sum_{b=j}^n e_{\sigma(b)} \right) A \quad (i+1 \leq j \leq n), & J_{n+1} &= 0, \\ F_j &= e_v J_j \quad (i+1 \leq j \leq n+1), \\ 0 &= F_{n+1} \subseteq F_n \subseteq \dots \subseteq F_{i+1} = K_v. \end{aligned} \quad (7.3)$$

For $i < j \leq n$, set $w = \sigma(j)$ and $L_{vw} = e_v C_w e_w$, a finite complex of vector spaces. Multiplication induces an isomorphism of dg right A -modules

$$L_{vw} \otimes_{\mathbb{k}} \Delta_w \xrightarrow{\sim} F_j / F_{j+1}, \quad p \otimes q \longmapsto pq. \quad (7.4)$$

Indeed, a basis path in F_j / F_{j+1} has greatest vertex w with respect to $\leq_\sigma$. Acyclicity permits a unique cut at its sole occurrence of w . The Leibniz rule makes the multiplication map in (7.4) a morphism of complexes. Splitting the finite complex L_{vw} over $\mathbb{k}$ yields

$$F_j / F_{j+1} \simeq \bigoplus_{h=1-M}^0 \Delta_w[-h]^{\oplus \dim_{\mathbb{k}} H^h(L_{vw})} \quad \text{in } \mathcal{D}_{\text{fd}}(A). \quad (7.5)$$

Step 3: admissibility of every filtration factor. If $H^h(L_{vw}) \neq 0$, a prefix path $p : v \rightarrow w$ of degree h occurs in L_{vw} . For every basis path q in Δ_w , the concatenation pq is a path, and $h + |q| \geq 1 - M$. Thus $\Delta_w[-h] \in \mathcal{H}_A^M$ and $0 \leq -h < M$. Every submodule and quotient in (7.3) is itself concentrated in $[1 - M, 0]$. The short exact sequences of the tower therefore induce conflations in $\mathcal{H}_A^M$. The direct sums in (7.5) refine the tower to a filtration by allowed shifted standards; a zero layer contributes no factors. Finally,

$$0 \longrightarrow K_v \longrightarrow P_v \longrightarrow \Delta_v \longrightarrow 0$$

is a short exact sequence of dg modules with all three terms in $\mathcal{H}_A^M$, so it induces the conflation required in condition dq2. For $m \geq M$, all the same objects and factors remain in $\mathcal{H}_A^m$, and every factor shift remains less than m . $\square$

For arrows concentrated in degree zero, $M = 1$ and $d_A = 0$. Theorem 7.1 therefore recovers quasi-heredity of $\mathbb{k}Q$ for every total order on the vertices.

Example 7.2 (Type A_2 : two orders and two kinds of standards). Let $A = \mathbb{k}(1 \xrightarrow{a} 2)$ with $|a| = -d$, $d \geq 1$, and $d_A = 0$. Take $m = d + 1$. Both total orders admit a structure, but their standards differ:

order	(Δ_1, Δ_2)	(K_1, K_2)	(∇_1, ∇_2)
$1 <_\sigma 2$	(S_1, S_2)	$(S_2[d], 0)$	$(S_1, \mathbb{D}(Ae_2))$
$2 <_\sigma 1$	(P_1, S_2)	$(0, 0)$	(S_1, S_2)

(7.6)

The costandard entries in (7.6) are labelled by vertices; their exceptional sequence is listed in decreasing order with respect to $\leq_\sigma$.

1. For $1 <_\sigma 2$, the presentation

$$S_2[d] \longrightarrow P_1 \longrightarrow S_1 \longrightarrow S_2[d + 1]$$

gives the exceptional sequence (S_1, S_2) . The shifted factor $S_2[d]$ is necessary when $d \geq 1$. The costandard $\nabla_2 = \mathbb{D}(Ae_2)$ has

$$H^q(\nabla_2) \cong \begin{cases} S_2, & q = 0, \\ S_1, & q = d, \\ 0, & q \notin \{0, d\}. \end{cases}$$

Thus the upper endpoint $m - 1 = d$ of the costandard window is attained.

2. For $2 <_\sigma 1$, the exceptional sequence is (S_2, P_1) . The standard P_1 has $H^{-d}(P_1) \cong S_2$, so negative cohomology in Δ_1 may contain S_2 , since $2 <_\sigma 1$. Both kernels vanish.

The costandard calculations follow directly from $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_u, S_v[t]) \cong H^t(S_v)e_u$ and $\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(S_u, \mathbb{D}(Ae_v)[t]) \cong \mathbb{D}(H^{-t}(S_u)e_v)$.

Example 7.3 (Type A_4 : standards with negative cohomology). Let $Q : 1 \xrightarrow{a} 2 \xrightarrow{b} 3 \xrightarrow{c} 4$, with $|a| = |b| = |c| = -1$ and $d_A = 0$. Then $M = 4$. Choose $(\sigma(1), \sigma(2), \sigma(3), \sigma(4)) = (2, 4, 1, 3)$, and put $U = P_1/(abA)$. The standards and kernels are

v	Δ_v	K_v	
1	U	$abA \cong \Delta_3[2]$	
2	S_2	$bA \cong \Delta_3[1]$	
3	P_3	0	
4	S_4	0	

 $\Delta^\sigma = (S_2, S_4, U, P_3).$
(7.7)

Here

$$H^{-1}(U) \cong S_2, \quad H^{-1}(P_3) \cong S_4, \quad 2 <_\sigma 1, \quad 4 <_\sigma 3.$$

The kernels K_1 and K_2 occupy degrees $[-3, -2]$ and $[-2, -1]$, respectively. Both kernels are single allowed factors in $\mathcal{H}_A^4$. The cohomology class of abc has degree -3 , so width 4 is already forced by the amplitude of A .

Example 7.4 (Type D_4 : a kernel filtered by shifts of two standards). Let Q have arrows $a : 1 \rightarrow 2$, $b : 2 \rightarrow 3$, and $c : 2 \rightarrow 4$, with $|a| = |c| = -1$, $|b| = 0$, and $d_A = 0$. Then $M = 3$. Choose $(\sigma(1), \sigma(2), \sigma(3), \sigma(4)) = (1, 3, 2, 4)$, and put $V = P_2/cA$. We obtain

$$\begin{array}{c|c|c} v & \Delta_v & K_v \\ \hline 1 & S_1 & aA \cong P_2[1] \\ 2 & V & cA \cong \Delta_4[1] \\ 3 & S_3 & 0 \\ 4 & S_4 & 0 \end{array} \quad \Delta^\sigma = (S_1, S_3, V, S_4). \quad (7.8)$$

The required standard filtration of K_1 is

$$0 \subset acA \subset aA, \quad acA \cong \Delta_4[2], \quad aA/acA \cong \Delta_2[1]. \quad (7.9)$$

Both factors lie in $\mathcal{H}_A^3$, and their vertex labels satisfy $1 <_\sigma 4$ and $1 <_\sigma 2$. The module V is concentrated in degree zero, whereas its shift $V[1]$ occurs as a kernel factor. Thus the projective expression $K_1 \cong P_2[1]$ must be refined as in (7.9) to verify condition **dq2**.

Example 7.5 (A non-trivial differential and an excluded width). Fix $d \geq 0$. Let Q have arrows $a : 1 \rightarrow 2$, $b : 2 \rightarrow 3$, and $c : 1 \rightarrow 3$, and define

$$|a| = 0, \quad |b| = -d, \quad |c| = -d - 1, \quad d_A(c) = ab, \quad d_A(a) = d_A(b) = 0. \quad (7.10)$$

The path-degree bound is $M = d + 2$. The pair $c \xrightarrow{d_A} ab$ is acyclic, so

$$H^*(A) \cong \mathbb{k}(1 \xrightarrow{a} 2 \xrightarrow{b} 3)/(ab) \quad \text{as graded algebras,} \quad |b| = -d. \quad (7.11)$$

The least integer m with $A \in \mathcal{H}_A^m$ is therefore $d + 1$. For the order $1 <_\sigma 3 <_\sigma 2$, the objects in (7.2) are

$$(\Delta_1, \Delta_2, \Delta_3) = (S_1, P_2, S_3), \quad K_1 = \mathbb{k}\{a, ab, c\}, \quad K_2 = K_3 = 0. \quad (7.12)$$

The only required kernel filtration is

$$0 \subset aA \subset K_1, \quad aA \cong \Delta_2, \quad K_1/aA \cong \Delta_3[d + 1]. \quad (7.13)$$

Consequently the order $1 <_\sigma 3 <_\sigma 2$ admits a structure at width $d + 2$. Nevertheless, $K_1 \simeq S_2$, and all the projectives, standards, and kernels in (7.12) already belong to $\mathcal{H}_A^{d+1}$.

The failure at width $d + 1$ is detected by the costandard ∇_3 . The triangles

$$\begin{array}{c} S_3[d] \longrightarrow P_2 \longrightarrow S_2 \longrightarrow S_3[d + 1], \\ S_2 \longrightarrow P_1 \longrightarrow S_1 \longrightarrow S_2[1] \end{array} \quad (7.14)$$

and the standard–costandard pairing give, for every $t \in \mathbb{Z}$,

$$\begin{aligned} H^t(\nabla_3)e_1 &= \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(P_1, \nabla_3[t]) \\ &\cong \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(S_2, \nabla_3[t]) \\ &\cong \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(S_3[d], \nabla_3[t - 1]) \\ &\cong \begin{cases} \mathbb{k}, & t = d + 1, \\ 0, & t \neq d + 1. \end{cases} \end{aligned} \quad (7.15)$$

In addition, $H^t(\nabla_3)e_2 = 0$ for every t , and $H^t(\nabla_3)e_3 \cong \mathbb{k}$ precisely when $t = 0$. Proposition 4.1 determines the standard objects from (A, σ) , independently of the width. Hence the costandard

window in Proposition 4.3 and (7.15) force $m \geq d + 2$ for every structure with the order $1 <_\sigma 3 <_\sigma 2$. The least width is exactly $d + 2$.

For $d = 0$, the algebra in (7.11) is ordinary, and the standard objects form a full exceptional sequence in $\mathcal{D}^b(\text{mod } H^0(A))$. Example 9.3 gives an elementary proof that (S_1, P_2, S_3) has no kernel filtration as required by condition dq2 at width 1. For $d \geq 1$, the same obstruction occurs with genuine negative cohomology in A . Thus cohomological amplitude alone does not determine the width of an m -qh structure, even for a smooth dg path algebra.

8 Excluded non-smooth algebras

The algebras in Examples 8.2, 8.3, 8.4, and 8.5 are not homologically smooth. Excluding those algebras is a consequence of conditions dq1 and dq2, and is a required feature of the definition.

Lemma 8.1 (Resolutions over graded dual numbers). *For an integer $\ell \geq 0$, set*

$$R_\ell = \mathbb{k} \oplus \mathbb{k}\varepsilon, \quad \varepsilon^2 = 0, \quad |\varepsilon| = -\ell, \quad d_{R_\ell} = 0, \quad S = R_\ell/(\varepsilon). \quad (8.1)$$

Then

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(R_\ell)}(S, S[t]) \cong \begin{cases} \mathbb{k}, & t \in (\ell + 1)\mathbb{Z}_{\geq 0}, \\ 0, & \text{otherwise.} \end{cases} \quad (8.2)$$

In particular, $S \notin \text{per } R_\ell$ and R_ℓ is not homologically smooth.

Proof. A semifree resolution $F \rightarrow S$ has

$$F = \bigoplus_{a \geq 0} x_a R_\ell, \quad |x_a| = -a(\ell + 1), \quad d_F(x_0) = 0, \quad d_F(x_a) = x_{a-1}\varepsilon \quad (a \geq 1). \quad (8.3)$$

The augmentation $F \rightarrow S$ sends x_0 to 1 and every x_a with $a > 0$ to zero. The relation $\varepsilon^2 = 0$ gives $d_F^2 = 0$. Each $x_a\varepsilon$ is the boundary of x_{a+1} , so the augmentation is a quasi-isomorphism. Applying the dg Hom complex from F to S gives zero differential and one basis vector in each degree $a(\ell + 1)$, proving (8.2). For a perfect object X , the total graded vector space $\bigoplus_t \text{Hom}_{\mathcal{D}_{\text{fd}}(R_\ell)}(X, S[t])$ is finite-dimensional. Thus (8.2) proves $S \notin \text{per } R_\ell$. Smoothness would imply that every object of $\mathcal{D}_{\text{fd}}(R_\ell)$ is perfect: tensor a finite construction of the diagonal bimodule with the object. Hence R_ℓ is not smooth. $\square$

Example 8.2 (Negative cohomology at the standard's own vertex). Fix $\ell \geq 1$, let $A = R_\ell$ be the algebra (8.1), and take $m = \ell + 1$. With one vertex, choose

$$\Delta_1 = P_1 = R_\ell, \quad K_1 = 0.$$

Condition dq2 holds. Every support restriction at a strictly larger vertex is vacuous, and

$$\text{End}_{\mathcal{H}_{R_\ell}^{\ell+1}}(\Delta_1) = \mathbb{k}, \quad [H^0(\Delta_1) : S] = 1.$$

Nevertheless,

$$H^{-\ell}(\Delta_1)e_1 = \mathbb{k}\varepsilon, \quad \text{Hom}_{\mathcal{D}_{\text{fd}}(R_\ell)}(\Delta_1, \Delta_1[-\ell]) = \mathbb{k}. \quad (8.4)$$

Thus allowing the standard's own simple in negative cohomology admits a non-smooth algebra, even when the standard is a brick and satisfies condition dq2. The vanishing of the first vector space in (8.4) is essential. More generally, for one vertex, condition dq2 forces $\Delta_1 \cong A$, and condition dq1 then forces $H^*(A) = \mathbb{k}$ in degree zero. Increasing m never makes R_ℓ satisfy both axioms.

Example 8.3 (Cohomological support does not replace the filtration). Take $A = R_1$, $m = 2$, and $\Delta_1 = S = A/(\varepsilon)$. The object Δ_1 satisfies condition **dq1**. In particular, Δ_1 is a brick. The short exact sequence

$$0 \longrightarrow S[1] \xrightarrow{\iota} A \xrightarrow{p} S \longrightarrow 0, \quad \iota(1) = \varepsilon, \quad (8.5)$$

induces a conflation in $\mathcal{H}_A^2$. There is no vertex $w >_\sigma 1$, so condition **dq2** requires $K_1 = 0$, whereas the kernel in (8.5) is $S[1] \neq 0$. If shifts of Δ_1 were admitted as factors of K_1 , then $K_1 = \Delta_1[1]$ would satisfy the weakened filtration requirement. Lemma 8.1 gives

$$\Delta_1 \notin \text{per } A, \quad \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_1, \Delta_1[2]) = \mathbb{k}.$$

Hence even condition **dq1** and an ordinary dg quotient $P_1 \twoheadrightarrow \Delta_1$ do not imply perfection, exceptionality, or smoothness. Requiring factors $\Delta_w[s]$ of K_v to satisfy $w >_\sigma v$ in condition **dq2** is indispensable.

Example 8.4 (The multiplicity-one condition in degree zero). Let $A = R_0 = \mathbb{k}[x]/(x^2)$ be concentrated in degree zero, take $m = 1$, and set $\Delta_1 = P_1 = A$, $K_1 = 0$. Then condition **dq2** holds and every off-diagonal support or orthogonality condition is vacuous. However,

$$[\Delta_1 : S_1] = 2, \quad \text{End}_A(\Delta_1) = A \not\cong \mathbb{k}.$$

Lemma 8.1 proves that A is not smooth. Thus the requirement $[H^0(\Delta_v) : S_v] = 1$ in condition **dq1** is necessary already for $m = 1$; off-diagonal projective tests and the kernel filtration alone do not recover classical quasi-heredity.

Example 8.5 (Exceptional terms and backward degree-zero Hom are insufficient). Let A be the dg algebra with arrows $a : 1 \rightarrow 2$ and $b : 2 \rightarrow 1$, relations $ab = ba = 0$, degrees $|a| = |b| = -1$, and zero differential. Paths are concatenated from left to right. Set $m = 2$, order $1 <_\sigma 2$, and

$$\Delta_1 = P_1, \quad \Delta_2 = P_2, \quad K_1 = K_2 = 0.$$

Condition **dq2** holds. Since $e_i A e_i = \mathbb{k}$, each Δ_i is exceptional. Moreover,

$$\text{Hom}_{\mathcal{H}_A^2}(\Delta_2, \Delta_1) = 0, \quad \text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(\Delta_2, \Delta_1[-1]) = \mathbb{k}a. \quad (8.6)$$

The cohomological support restriction in condition **dq1** excludes $H^{-1}(\Delta_1)e_2 = \mathbb{k}a$.

To verify non-smoothness directly, let $F \rightarrow S_1$ be the semifree resolution with a generator x_j in degree $-2j$, at vertex 1 for even j and vertex 2 for odd j . Define

$$d_F(x_0) = 0, \quad d_F(x_j) = \begin{cases} x_{j-1}a, & j \text{ odd}, \\ x_{j-1}b, & j > 0 \text{ even}. \end{cases} \quad (8.7)$$

The augmentation sends x_0 to a generator of S_1 and every x_j with $j > 0$ to zero. The relations give $d_F^2 = 0$; every radical basis vector is the boundary of the next generator. Thus $F \rightarrow S_1$ is a quasi-isomorphism, and

$$\text{Hom}_{\mathcal{D}_{\text{fd}}(A)}(S_1, S_1[t]) \cong \begin{cases} \mathbb{k}, & t \in 4\mathbb{Z}_{\geq 0}, \\ 0, & \text{otherwise}. \end{cases} \quad (8.8)$$

Therefore $S_1 \notin \text{per } A$ and A is not smooth. In particular, adding individual exceptionality to degree-zero backward orthogonality still does not protect smoothness.

9 Rejected standards and orders on smooth algebras

Failure of chosen standards to satisfy the axioms need not indicate a singularity of the algebra. Examples 9.1 and 9.2 distinguish a rejected choice of standards from an order for which no choice can satisfy the axioms.

Example 9.1 (Positive shifts of the target detect only H^0). For $d \in \{1, 2\}$, let

$$A_d = \mathbb{k}(1 \xrightarrow{a} 3 \xleftarrow{b} 2), \quad |a| = -d, \quad |b| = -1, \quad d_{A_d} = 0, \quad m = d + 1, \quad (9.1)$$

and use the order $1 <_\sigma 2 <_\sigma 3$. Consider

$$(\Delta_1, \Delta_2, \Delta_3) = (P_1, S_2, P_3), \quad (K_1, K_2, K_3) = (0, P_3[1], 0). \quad (9.2)$$

The only non-identity presentation is induced by

$$0 \longrightarrow P_3[1] \xrightarrow{\iota_b} P_2 \longrightarrow S_2 \longrightarrow 0, \quad \iota_b(e_3) = b. \quad (9.3)$$

All three objects in (9.3) belong to $\mathcal{H}_{A_d}^{d+1}$, and $K_2 = \Delta_3[1]$ is an allowed factor. Thus condition dq2 holds. Each Δ_i is exceptional, and $H^0(\Delta_i) = S_i$. Consequently,

$$\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A_d)}(P_w, \Delta_v[k]) = H^k(\Delta_v)e_w = 0 \quad (v <_\sigma w, 0 \leq k < m). \quad (9.4)$$

For any standard in $\mathcal{H}_A^m$, every test with $k > 0$ in (9.4) vanishes automatically. Writing $\mathrm{Hom}_{\mathcal{H}_A^m}(P_w, \Delta_v[k])$ instead does not change the calculation when $\Delta_v[k] \in \mathcal{H}_A^m$; the latter Hom space is undefined when $\Delta_v[k] \notin \mathcal{H}_A^m$.

The actual backward morphism spaces for (9.2) are

$$\begin{aligned} \mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A_d)}(\Delta_2, \Delta_1[t]) &= \begin{cases} \mathbb{k}, & t = 2 - d, \\ 0, & t \neq 2 - d, \end{cases} \\ \mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A_d)}(\Delta_3, \Delta_1[t]) &= \begin{cases} \mathbb{k}, & t = -d, \\ 0, & t \neq -d, \end{cases} \\ \mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A_d)}(\Delta_3, \Delta_2[t]) &= 0. \end{aligned} \quad (9.5)$$

Indeed, $\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A_d)}(P_2, P_1[t]) = 0$ for all t , so (9.3) gives

$$\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A_d)}(S_2, P_1[t]) \cong \mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A_d)}(P_3[1], P_1[t-1]) \cong H^{t-2}(P_1)e_3.$$

The other two equalities in (9.5) are projective calculations. Thus:

- (i) for $d = 1$, all backward degree-zero Hom spaces vanish, but $\mathrm{Hom}_{\mathcal{D}_{\mathrm{fd}}(A_1)}(\Delta_2, \Delta_1[1]) = \mathbb{k}$;
- (ii) for $d = 2$, even backward degree-zero orthogonality fails: $\mathrm{Hom}_{\mathcal{H}_{A_2}^3}(\Delta_2, \Delta_1) = \mathbb{k}$.

For both values of d , condition dq1 rejects the chosen $\Delta_1 = P_1$, since $H^{-d}(P_1)e_3 = \mathbb{k}a$. The algebra A_d does admit an m -qh structure for the same order: choose $\Delta'_i = S_i$ for every i , with

$$K'_1 = P_3[d], \quad K'_2 = P_3[1], \quad K'_3 = 0.$$

Conditions dq1 and dq2 hold for $(\Delta'_1, \Delta'_2, \Delta'_3)$, so A_d is smooth. The failure in (9.2) concerns the chosen standards, rather than the existence of an m -qh structure.

Example 9.2 (A smooth algebra with an order excluded at every width). Let A be the ordinary algebra with arrows $a : 1 \rightarrow 2$ and $b : 2 \rightarrow 1$, and relation $ab = 0$. A basis of A is e_1, e_2, a, b, ba , with paths concatenated from left to right. The right projectives satisfy

$$\mathrm{rad} P_1 = aA \cong S_2, \quad \mathrm{rad} P_2 = bA \cong P_1. \quad (9.6)$$

Thus $\text{pdim}_A S_2 = 1$ and $\text{pdim}_A S_1 = 2$, whence $\text{gldim } A = 2$. In particular, A is homologically smooth.

Take the order $1 <_\sigma 2$. For every $m \geq 1$, the greatest vertex is 2, so condition **dq2** forces $\Delta_2 \cong P_2$. But

$$H^0(P_2)e_2 = e_2 A e_2 = \mathbb{k}e_2 \oplus \mathbb{k}ba, \quad (9.7)$$

contradicting $\dim_{\mathbb{k}} H^0(\Delta_2)e_2 = 1$ in condition **dq1**. Therefore no m -qh structure exists for the order $1 <_\sigma 2$, at any width.

For the opposite order $2 <_\sigma 1$, choose

$$\Delta_1 = P_1, \quad \Delta_2 = S_2, \quad K_1 = 0, \quad K_2 = bA \cong \Delta_1.$$

Equations (9.6) show that both axioms hold already for $m = 1$. Hence increasing the width does not remove every obstruction arising from the choice of order.

Example 9.3 (Full exceptionality does not imply the kernel filtration). Consider the ordinary algebra

$$B = \mathbb{k}(1 \xrightarrow{a} 2 \xrightarrow{b} 3)/(ab), \quad 1 <_\sigma 3 <_\sigma 2, \quad (9.8)$$

and the quotient modules

$$(\Delta_1, \Delta_2, \Delta_3) = (S_1, P_2, S_3), \quad (K_1, K_2, K_3) = (S_2, 0, 0). \quad (9.9)$$

The standards satisfy condition **dq1**. The sequence $(\Delta_1, \Delta_3, \Delta_2) = (S_1, S_3, P_2)$ is a full exceptional sequence in $\mathcal{D}_{\text{fd}}(B)$:

- (i) the projectives $S_3 = P_3$ and P_2 are exceptional;
- (ii) the resolution $0 \rightarrow P_3 \rightarrow P_2 \rightarrow P_1 \rightarrow S_1 \rightarrow 0$ makes S_1 exceptional;
- (iii) all backward derived Hom spaces vanish, since their sources are P_2 or P_3 , and the corresponding vertex components of their targets vanish;
- (iv) the triangle $S_3 \rightarrow P_2 \rightarrow S_2 \rightarrow S_3[1]$ puts S_2 in the thick subcategory generated by S_3, P_2 , so the standards generate every simple and hence $\mathcal{D}_{\text{fd}}(B)$.

Nevertheless, condition **dq2** fails for $m = 1$. A filtration of $K_1 = S_2$ in $\text{mod } B$ by $\Delta_2 = P_2$ and $\Delta_3 = S_3$, whose labels satisfy $2 >_\sigma 1$ and $3 >_\sigma 1$, would give

$$[S_2] = x[P_2] + y[S_3] = x[S_2] + (x + y)[S_3], \quad x, y \in \mathbb{Z}_{\geq 0}, \quad (9.10)$$

in $K_0(\text{mod } B)$, forcing $x = 1$, $y = -1$. Thus even a full exceptional sequence satisfying condition **dq1**, together with quotient presentations $P_v \twoheadrightarrow \Delta_v$, does not imply condition **dq2**. In fact, no width-1 structure exists for the order in (9.8): condition **dq2** forces $\Delta_2 = P_2$, while condition **dq1** and the quotient presentations force $\Delta_3 = S_3$ and $\Delta_1 = S_1$.

For $m = 2$, the same ordinary algebra and the same standards do satisfy condition **dq2**. Rotate the short exact sequence $0 \rightarrow S_3 \rightarrow P_2 \rightarrow S_2 \rightarrow 0$ to obtain the conflation

$$P_2 \longrightarrow S_2 \longrightarrow S_3[1] \dashrightarrow \quad \text{in } \mathcal{H}_B^2. \quad (9.11)$$

The two-factor filtration $0 \rightarrow P_2 \rightarrow S_2$ has factors Δ_2 and $\Delta_3[1]$. The vertex labels of both factors satisfy $2 >_\sigma 1$ and $3 >_\sigma 1$, so $K_1 \in \text{Filt}_{\mathcal{H}_B^2} \{\Delta_2, \Delta_3[1]\}$. The map $P_2 \rightarrow S_2$ in (9.11) is an inflation in $\mathcal{H}_B^2$, although it is a surjection of ordinary modules. Thus an extriangulated filtration must not be replaced by a chain of dg submodules. Example 7.5 gives a path dga with non-trivial differential realising the same distinction between widths 1 and 2.

10 Summary of consequences and counterexamples

Conditions **dq1** and **dq2** have the consequences in Table 1. The negative examples in Table 2 identify failed weakenings of the axioms and genuine restrictions on orders and widths.

Consequence of (dq1) and (dq2)	Precise result
Every standard is a brick and exceptional	Theorem 3.2.
Backward orthogonality in every degree	Equation (3.5).
Standards are full in $\mathcal{D}_{\text{fd}}(A) = \text{per } A$	Theorems 2.2 and 3.2.
Homological smoothness	A theorem, without an additional axiom; Theorem 2.2.
Canonical standard objects	Determined by (A, σ) ; Proposition 4.1.
Costandards $\nabla_v \in \mathcal{H}_A^m[1 - m]$	Proposition 4.3.
Recognition of allowed filtrations	Proposition 5.1.
Least width and all larger widths	Corollary 5.3.
Classical quasi-heredity at $m = 1$	Corollary 3.4.
Injective presentations and passage to A^{op}	Proposition 6.2.
Euler Cartan determinant 1	Proposition 6.1.
All vertex orders for graded acyclic path dgas	At the path-degree bound; Theorem 7.1.

Table 1: Properties proved from the two axioms.

Claim disproved	Counterexample and reason
The restrictions $H^q(\Delta_v)e_v = 0$ for $q < 0$ can be omitted	Example 8.2: graded dual numbers are non-smooth despite a brick projective standard.
Composition factors and a quotient presentation suffice	Example 8.3: the simple over dg dual numbers satisfies (dq1), but its kernel is a shift of the same standard.
The requirement $[H^0(\Delta_v) : S_v] = 1$ is dispensable	Example 8.4: ordinary dual numbers.
Exceptional terms and backward degree-zero Hom protect smoothness	Example 8.5: a graded radical-square-zero cycle.
Positive target shifts test the required support	Example 9.1: the tests miss backward morphisms in degree 1 or degree 0.
Smoothness permits every vertex order at some width	Example 9.2: the greatest vertex is 2, but $\dim_{\mathbb{k}} H^0(P_2)e_2 = 2$.
A full exceptional sequence implies (dq2)	Example 9.3: a required classical filtration would have a negative multiplicity.
The least m with $A \in \mathcal{H}_A^m$ suffices	Example 7.5: a required costandard degree forces a larger width.
All filtrations have invariant unsigned multiplicities	Example 5.2: zero has a two-factor filtration, even over a semisimple algebra.
Extriangulated filtrations are chains of submodules	Equation (9.11): a surjection of modules is an inflation in $\mathcal{H}_B^2$.

Table 2: Failures of weakened axioms or stronger classical expectations. No non-smooth algebra satisfies both (dq1) and (dq2).

References

- [BB26] A. Bodzenta and A. Bondal, *Highest weight categories via pairs of dual exceptional sequences*, 2026. [arXiv:2601.22004](#). Cited on page 5.
- [MP25] N. Mochizuki and M. Plogmann, *On the Auslander–Reiten theory for extended hearts of proper connective dg algebras*, 2025. [arXiv:2505.16560](#). Cited on page 2.
- [RS22] T. Raedschelders and G. Stevenson, *Proper connective differential graded algebras and their geometric realizations*, *European J. Math.* **8** (2022), S574–S598. [arXiv:1903.02849](#). Cited on page 3.